

Module 3

Using Formula Fields & Formula Editor

Formula Fields

- Formula fields are custom fields which generate the results dynamically based on the calculations/expressions.
- Formula Fields are Read-only.
- Can refer standard fields, custom fields, and other formula fields.
- Can refer fields from the same object or related parent objects.
- It automatically updates in real time whenever record is accessed.
- Formula field can override security. If it contains any sensitive information we need to use field level security.
- We cannot delete field directly which is referenced in formula field expression.

Types of Formula Fields

Simple Formula Field:

A Formula field which performs basic/simple calculations or manipulation on data like concatenating two text fields, manipulating dates etc. It displays the fields of the same object.

Advanced Formula:

Advanced Formula can be used to implement complex calculations which includes functions, Text Manipulations, Conditional Logics.

- It can perform mathematical calculations, aggregate data from related records.
- It can manipulate text strings, extract substrings, format date and time.
- It can include conditional statements such as IF THEN ELSE, AND, OR, Nested Functions.
- It can create Hyperlinks and display images.

3.1 Using Formulas and Formula Editor

1. Create Formula Field on Position Object to calculate number of days the position has been open.
2. Create a formula field on Position object to show Demand Level based on Salary Range
If **Salary Range** is **120k+** display as **High** and If **Salary Range** is **40k-60k** display as **Low** else **Medium**.

Using Formula Fields & Formula Editor

Important Text Functions

Functions	Explanation
LEFT(text,num_chars)	Returns specified number of characters from left. text : The text string from which you want to extract characters. num_chars : The number of characters to return
LEN(text)	Returns Length of string text : The text string whose length you want to determine.
TRIM(text)	Removes leading and trailing spaces from text
SUBSTITUTE(text, old_text, new_text)	Replace specific string with new text text : text string containing the substring you want to replace. old_text : The substring you want to replace. new_text :The substring to replace old_text with.

Functions	Explanation
ISPICKVAL(picklist_field, text_literal)	Checks if a picklist field's value matches a specified string literal. picklist_field : value of the picklist field text_literal : The text value which you want to compare.
CONTAINS(text, compare_text)	Checks if a text includes specific characters and returns TRUE if it contains else FALSE. text : the string to search within compare_text : The substring to search for within the text string.
UPPER(text)	Converts all letters in the text to Upper case.

Functions	Explanation
TODAY()	Returns the current date
DATE(year,month,day)	Creates a date from a year, month and day
YEAR(date)	Returns Year of a date
DAY(date)	Returns Day of a date
DATEVALUE()	Converts a text string to DATE
ADDMONTHS(date,num)	Adds a specified number of months to a date

Functions	Explanation
IF(conditional statement, value_if_true, value_if_false)	Checks whether a condition is true,and returns one value if true and another value if false.
ISBLANK(expression)	Checks whether expression is blank and returns True or false
AND(logical1,logical2,...)	Returns True if all arguments are true otherwise returns False
NOT(logical)	Changes FALSE to TRUE or TRUE to FALSE
CASE(expression, value1, result1, value2, result2,...,else_result)	Compares an expression with several values. If the expression matches any value, it returns the corresponding result. If it doesn't match any values, it returns a different result.

Functions	Explanation
IMAGE(image_url, alternate_text [, height, width])	Displays an image from a specified URL, with an optional alternate text. Additionally, it can include parameters for height and width.
HYPERLINK(url, friendly_name [, target])	Creates a hyperlink with a specified URL and display text, optionally specifying the target window or frame.

Cross Object Formula Field

- Cross object formula field can be created in object and can reference fields from related objects in Master detail Relationship/Lookup Relationship.
- It allows us to automatically calculate and display data from multiple related objects within a single field.
- It allows us to perform all operations like arithmetic, logical and text-based operations on referenced data.
- It can reference fields from objects up to 10 relationships away

Limitations/Restriction of Formula Field

- Formula fields cannot exceed 3,900 characters including spaces , return characters and comments.
- We cannot use long text area, Description fields in formula
- We cannot delete the fields directly which are referenced in formula.
- Complex calculation can impact page load time.

Implementing Roll-up Summary Field

Introduction to Rollup Summary Fields

- It is a read only field. It summarizes the values of fields in detail records and displays the aggregated result in Master object.
- It performs calculation such as SUM, COUNT, MIN, MAX on child records related to a master record.
- We can include all the records in the calculation, or we can filter records with a certain criteria using filters.

Defining Rollup Summary Fields

Operations Allowed on Datatypes :

- Sum - Number , Currency & percent
- Max / Min - Number , Currency , percent ,Date, Date & Time
- We can create 25 Rollup summary fields per object can be extended to 40.
- Rollup summary fields are only allowed in Master detail Relationship, and it is not available in Lookup Relationship.

Exceptions on Standard objects :

There are some exceptions in standard objects. Even though there is no M-D relationship we would be able to create Rollup summary fields in below objects.

- Account and Opportunity
- Opportunity and Opportunity Products
- Order and Order Products
- Campaign and Campaign Members

These objects have Lookup Relationship.

Validation Rules

Introduction to Validation Rules

Validation rules enforce specified standards or conditions on data entered by user. It prevents the record from being saved until those criteria are met.

Types of Validations :

1. System Validations
2. Custom Validations

System Validations:

System validations are enforced by salesforce which comes by default.

Ex : Datatype validations for all fields like Number field cannot contain alphabets ,
phone number should be in valid format.

Email address must be in valid format and universally required fields cannot be empty

Custom Validation Rules

- We can create our own validations to ensure data entered by user in valid format.
- Verifies the data entered by the user into record meets the specified standards before saving record.
- It contains Formula/Expression/Error Condition which evaluates the entered data and returns value **true** or **false**.
- When the expression returns **true** then it displays error message and if it returns **false** then record gets saved.
- We can clone the existing validation rule and can create new rule.
- We can deactivate the validation rule when needed by unchecking the Active checkbox.

Custom Validation Rules

Error Message:

We can specify what error message needs to be displayed and the location to be displayed.

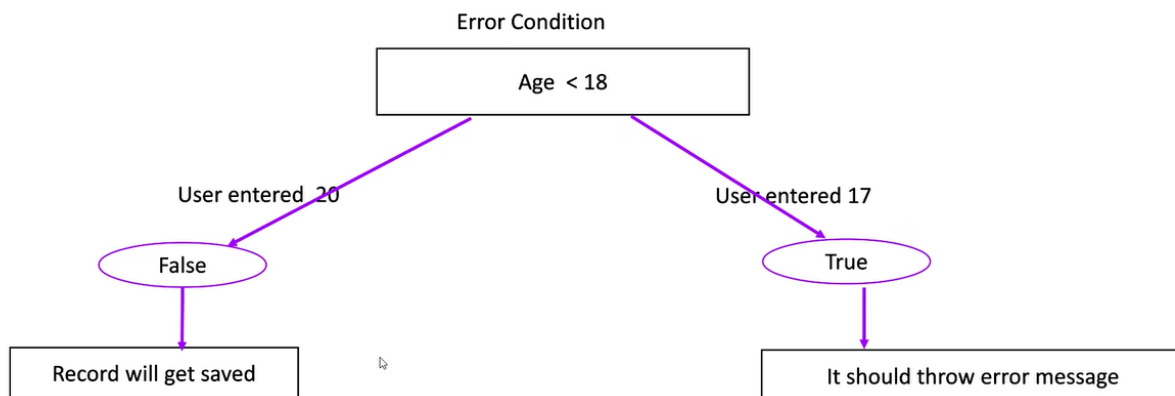
Top of the Page :

When we are creating validation rule on multiple fields together. We can display error message at top of the page as a popup that covers whole width.

On the Field:

Error message is displayed directly adjacent to field.

- Condition: Age should be greater than 18 . If user enters age which is less than 18, it should throw error message



Important Functions

Functions	Explanation
ISCHANGED(field)	Compares the value of a field to the previous value and returns TRUE if the values are different. If the values are same, then returns FALSE.
ISNEW()	Checks if the record is new and returns TRUE if it is new. Otherwise, returns FALSE
PRIORVALUE(field)	Returns previous value of the field.
LEN(string)	Returns the length of the string.
REGEX(text,regex_pattern)	Compares a text field to a regular expression and returns TRUE if there is a match Otherwise, returns FALSE.

Debug Logs

What are debug logs

- Debug logs are valuable tool for troubleshooting issues, monitoring system behavior and analyzing the code execution. It can capture information about database operations, Apex code errors and resources.
- We can set expiration date up to 24 hours only.

Debug Log Levels

Developers can specify log levels to indicate the level of detail to capture.

Levels include:

Error, Warn, Info, Debug, Fine, Finer, Finest

New Debug Level Help for this Page

Debug Level Details Cancel Save Reset

Name

The Name field can only contain underscores and alphanumeric characters. It must be unique, begin with a letter, not include spaces, not end with an underscore, and not contain two consecutive underscores.

For each category, such as Workflow or Apex Code, use the level to specify the amount of information to be recorded in the debug log. The combination of category and level specify which events get logged. Changes to the filter only apply to future logged events.

Category	Level	Events
Database	Info	SOQL_EXECUTE_BEGIN, SOQL_EXECUTE_END, SOSL_EXECUTE_BEGIN, SOSL_EXECUTE_END, QUERY_MORE_ITERATIONS, DML_BEGIN, DML_END, FOR_UPDATE_LOCKS_RELEASE, BULK_DML_RETRY, SAVEPOINT_SET, SAVEPOINT_ROLLBACK, SAVEPOINT_RESET, SAVEPOINT_RELEASE
Workflow	None	WF_RULE_INVOCATION, WF_APPROVAL, WF_FIELD_UPDATE, WF_SPOOL_ACTION_BEGIN, WF_ACTION, WF_FORMULA, WF_RULE_EVAL_BEGIN, WF_RULE_EVAL_END, WF_RULE_EVAL_VALUE, WF_CRITERIA_BEGIN, WF_CRITERIA_END, WF_RULE_ENTRY_ORDER, WF_RULE_NOT_EVALUATED, WF_RULE_FILTER, WF_ESCALATION, WF_ESCALATION_ACTION, WF_TIME_TRIGGER, WF_ACTIONS_BEGIN, WF_TIME_TRIGGER, WF_ACTIONS_END, WF_ENQUEUE_ACTIONS, WF_APPROVAL_SUBMIT, WF_APPROVAL_SUBMITTER, WF_APPROVAL_REMOVE, WF_NEXT_APPROVER, WF_EVAL_ENTRY_CRITERIA, WF_PROCESS_FOUND, WF_SOFT_REJECT, WF_HARD_REJECT, WF_PROCESS_NODE, WF_ASSIGN, WF_REASSIGN, WF_RESPONSE_NOTIFY, WF_OUTBOUND_MSG, WF_ACTION_TASK, WF_EMAIL_ALERT, WF_EMAIL_SENT, SLA_PROCESS_CASE, SLA_NULL_START_DATE, SLA_EVAL_MILESTONE, SLA_END, WF_KNOWLEDGE_ACTION, WF_SEND_ACTION, WF_CHATPOST, WF_QUICK_CREATE, WF_FLOW_ACTION_BEGIN, WF_FLOW_ACTION_END, WF_FLOW_ACTION_ERROR, WF_FLOW_ACTION_ERROR_DETAIL, WF_APEX_ACTION_EVENT, SERVICE_PUB_BEGIN, EVENT_SERVICE_PUB_END, EVENT_SERVICE_SUB_BEGIN, EVENT_SERVICE_SUB_END, SCHEDULED_FLOW_DETAIL, FLOW_CREATE_INTERVIEW_BEGIN, FLOW_CREATE_INTERVIEW_END, FLOW_CREATE_INTERVIEW_ERROR, FLOW_START_INTERVIEWS_BEGIN, FLOW_START_INTERVIEWS_END, FLOW_START_INTERVIEWS_ERROR, FLOW_ELEMENT_FAULT, FLOW_ELEMENT_ERROR, FLOW_BULK_ELEMENT_NOT_SUPPORTED, INVOCABLE_ACTION_ERROR, ROUTE_WORK_ACTION_ADD, SKILL_REQUIREMENT_ACTION_ADD, SCREEN_POP_ACTION, VALIDATION_RULE, VALIDATION_FAIL, VALIDATION_PASS, VALIDATION_ERROR, VALIDATION_FORMULA
Validation	Info	CALLOUT_REQUEST, CALLOUT_RESPONSE, FUNCTION_INVOCATION, REQUEST_FUNCTION_INVOCATION_RESPONSE, XDS_RESPONSE, XDS_RESPONSE_ERROR, NAMED_CREDENTIAL_REQUEST, NAMED_CREDENTIAL_RESPONSE, EXTERNAL_SERVICE_REQUEST, EXTERNAL_SERVICE_RESPONSE, EXTERNAL_SERVICE_CALLBACK
Callouts	Info	

Debug log details

Enable Debug logs:

Navigate to Setup > Debug Logs.

- Click on New to create a new debug log configuration.
- Select the user for whom you want to generate debug logs.
- Define the log levels for various categories such as Validation Rules, database etc.
- Save the configuration.

View/Download Debug Logs:

- You can view as well as download debug logs for further analysis.
- Debug logs contain detailed information about the execution of code, including debug statements, error messages, database operations.